Preprint

Simulation-based inference for cultural evolution: a tutorial with baby name data

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ABSTRACT

Cultural evolutionary models are usually easy to simulate but difficult to fit, because most have no tractable likelihood. Simulation-based inference (SBI) addresses this by treating the simulator itself as the likelihood, and now ranges from rejection-based approximate Bayesian computation (ABC) to amortized neural posterior estimation (NPE). Knowing which to use is hard, partly because they are rarely compared on the same model and data. In this tutorial I fit a Wright–Fisher-style model of baby name popularity to uncensored female naming records from South Australia (1944–2017, with inference on 1981–2017), comparing rejection ABC, ABC with random forests, and NPE with BayesFlow on one shared set of 10,000 simulations. The model includes frequency dependence (β), an age-dependent bias (α), and random variation (γ). Recovery varies widely across parameters and methods, from $r = 0.96$–1.00 for β to $r = 0.39$–0.97 for γ, and these differences track what each estimator does with the summary statistics it is given and, for γ, how much of the signal those statistics retain. I use the comparison to work through what makes a parameter identifiable, the diagnostics that indicate whether a posterior can be trusted, and common failure modes. All code and data are provided.

Keywords: simulation-based inference; approximate Bayesian computation; neural posterior estimation; cultural evolution; summary statistics

Social media summary: Three simulation-based inference methods compared on one cultural evolution model of baby name popularity.

1 INTRODUCTION

Cultural evolution combines insights from evolutionary biology and cognitive science to understand how learned behaviors and ideas change over space and time. Simulations have played a core role in the growth of the field, letting researchers explore how individual-level mechanisms may generate population-level patterns (Acerbi et al., 2022). Connecting those simulations to real data, though, remains difficult.

The difficulty is specific. Most statistical inference requires a likelihood, or a function $p(x \mid \theta)$ that returns the probability of the data x under a set of parameters θ. When one is available, a large body of machinery (e.g., maximum likelihood, MCMC, variational inference) can be brought to bear. Cultural evolutionary models rarely provide one. We can run them forward to generate trajectories and turnover statistics, but we usually cannot write down the probability of any particular outcome (Kandler & Powell, 2018). In the model used here, for example, every name receives a random shock each year that is never observed, so computing the probability of the data would require summing over every possible combination of shocks across the thousands of names active in each year, across all years.

Simulation-based inference (SBI) links simulations to data, and is an increasingly important part of the cultural evolutionary toolkit (Cranmer et al., 2020; Csilléry et al., 2010; Kandler & Powell, 2018; Marin et al., 2012). The core move is to replace evaluation of the likelihood with sampling from it: if a model can be simulated, the relationship between parameters and data can be learned from the simulations themselves and then applied to the real data. Over the last two decades SBI has been applied to bird and whale song (Lachlan et al., 2018; Pitocchelli et al., 2025; Youngblood & Lahti, 2022; Zandberg et al., 2021), social media use (Carrignon et al., 2019; Youngblood et al., 2023), and other aspects of human behavior (Carrignon et al., 2020; Deffner et al., 2024; Youngblood, 2019; Youngblood & Passmore, 2026).

SBI is a family of methods rather than a single one, though, and that family has grown quickly. Classical approximate Bayesian computation (ABC), including rejection sampling and regression adjustment, remains widely used in R-centric workflows (Beaumont, 2010; Raynal et al., 2019; Sisson et al., 2018). Neural posterior estimation (NPE) has more recently emerged as a scalable alternative that amortizes inference across observations after a single training phase (Radev et al., 2022; 2023), so training cost is paid once and then spread across every dataset the network is later applied to. It also brings the inferential power of modern deep learning to bear, which makes it possible to fit more complex models with more parameters than classical ABC can handle. These approaches are seldom compared on the same model with the same simulations, so a researcher choosing between them has relatively little to go on. This tutorial provides that comparison, and uses it to illustrate the strengths and weaknesses of each method.

The example we will use is a compact model of baby name popularity fit to uncensored South Australian records. Baby names are one of the classic model systems of cultural evolution, for three reasons. First, national registers record the name of essentially every child born, so the data are close to a complete census of a cultural trait rather than a sample, and run over a century or more. Second, names are transmitted socially but carry almost no practical function, so differences in their success are difficult to attribute to any material advantage of one variant over another. That makes naming an unusually clean place to test whether observed change can be explained by unbiased copying alone. Third, names turn over quickly enough that the dynamics are visible within a few decades.

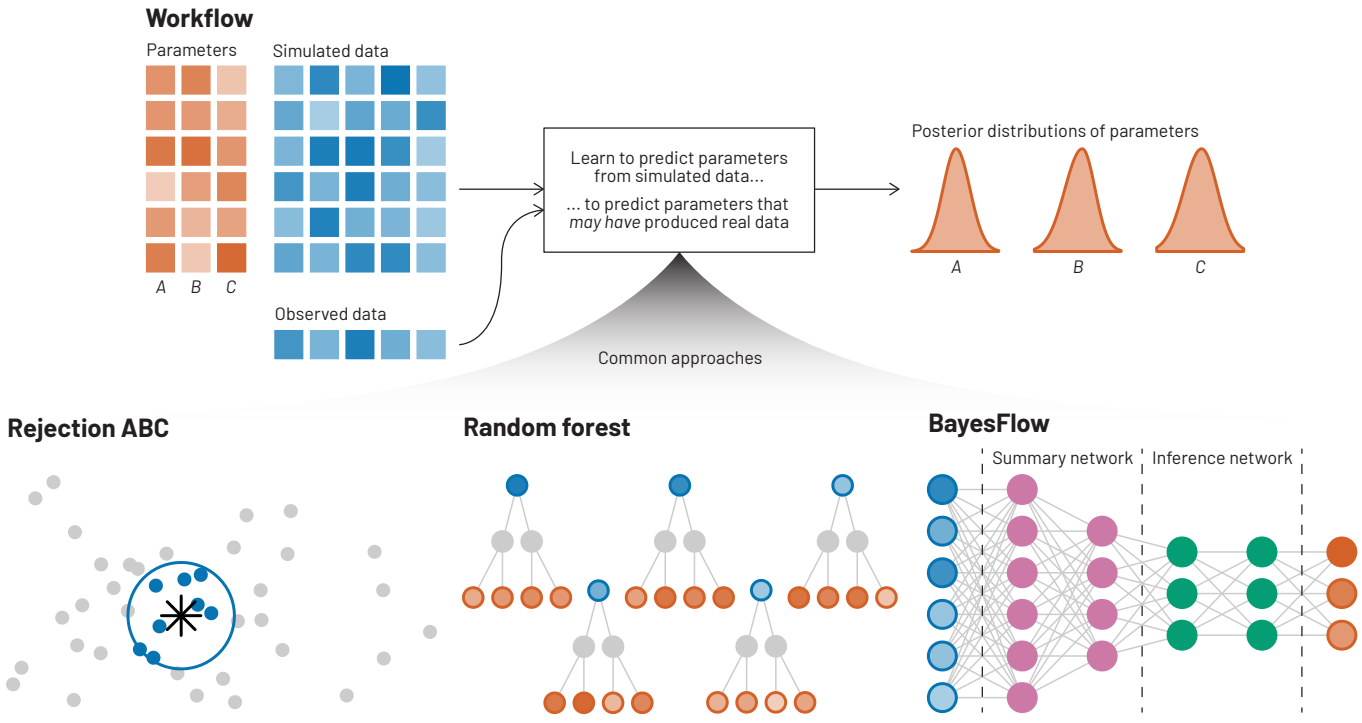


Figure 1: Shared SBI pipeline and three implementations. Simulated parameter–data pairs are used to learn a mapping from summary statistics to parameters; observed data then enter the same mapping to produce posteriors. The three methods open that mapping differently: rejection ABC retains simulations in an acceptance neighborhood of the observed statistics, ABC-RF fits an ensemble of regression trees, and BayesFlow trains a summary network plus an inference network on the full time series.

The literature reflects this. (Hahn & Bentley, 2003) introduced baby names as a test case for drift, showing that the frequency distribution of American names is broadly consistent with random copying; related work extended the argument to turnover and other domains of popular culture (Bentley et al., 2004; 2007). Subsequent studies have asked where that account breaks down. Turnover in the top-k names has been used to detect departures from neutrality (Acerbi & Bentley, 2014), rare variants carry much of the information that distinguishes neutrality from a novelty bias (O'Dwyer & Kandler, 2017), and non-equilibrium neutral models have been developed for cultural frequency data more generally, since populations are rarely at the equilibrium neutral theory assumes (Kandler & Shennan, 2013). The clearest departure comes from (Newberry & Plotkin, 2022), who fit a Wright–Fisher model with a frequency-dependent selection coefficient to annual naming records from the United States, France, Norway and the Netherlands. All four show negative frequency dependence: the most common names decline by 2–6% per year, names as rare as 1 in 10,000 births grow by 1–3% per year, and the changeover between growth and decline falls between 8 and 32 births per 10,000. Others have asked how a name's popularity feeds back on its own adoption (Berger & Le Mens, 2009) and how individual naming decisions aggregate into collective patterns (Gureckis & Goldstone, 2009), anticipated in Lieberson's account of naming as fashion without advertising (Lieberson, 2000). The model used here is closest to these frequency-dependent models, with an age term and a random-variation term added.

The core notation is collected in Table 1, and symbols local to a single equation are defined where they appear.

2 THE INFERENCE PROBLEM

2.1 Replacing the likelihood with simulation

Two terms recur from here on. The prior, $p(\theta)$, is the distribution we draw parameter values from before looking at the data: it says which values are on the table, and how plausible each is in advance. The posterior, $p(\theta \mid x)$, is what we want at the end: the values that remain plausible once the data x are taken into account. Everything in this tutorial is a way of getting from the first to the second without a likelihood.

Sampling from $p(x \mid \theta)$ is straightforward, since that is what the simulator does, so we generate many parameter–data pairs and learn the inverse mapping from them:

$$\theta^{(m)} \sim p(\theta) \qquad \text{draw one parameter set from the prior}$$
$$x^{(m)} \sim p\left(\cdot \mid \theta^{(m)}\right) \qquad \text{run the simulator at those values to get one record}$$

The two steps are repeated for $m = 1, \ldots, M$, and the superscript is simply a counter: $\theta^{(7)}$ and $x^{(7)}$ are the parameters and the record from the seventh run. What we end up with is a table of M rows, each pairing chosen parameter values with the data they produced.

This archive of M pairs is the raw material for every method used here, and the three differ only in what they do with it (Figure 1). Running the simulations once and sharing them across methods keeps the comparison fair.

Table 1: Notation used throughout.

Symbol	Meaning
$\theta = (\beta, \alpha, \gamma)$	Model parameters
β	Frequency dependence ($\beta > 1$ is conformity, $\beta < 1$ anticonformity)
α	Age-dependent bias ($\alpha > 0$ favors young names, $\alpha < 0$ old names)
γ	Standard deviation of the log-normal random variation in weights (`noise_sd` in the code)
n_t	Vector of name counts in year t
N_t	Total births in year t
x	Full simulated or observed record, $(n_1, \ldots, n_T)$
$S(x)$	Summary statistics (six per year, or their medians)
S_{obs}	Summary statistics of the observed record
$T = 37$	Number of simulated years (1981–2017)
$M = 10^4$	Number of prior simulations
ε	ABC tolerance, or acceptance radius
$d(\cdot, \cdot)$	Euclidean distance on z-scored summary statistics

2.2 Three approximations

Rejection ABC targets an approximation to the posterior rather than the posterior itself, and it is easiest to build in two steps. The first step asks how often a given parameter value works. Run the simulator once at θ, summarize what comes out, and check whether it lands close enough to the observation to keep; the probability of that happening is

$$A(\theta) = \Pr[d(S(x), S_{\text{obs}}) \le \varepsilon], \quad x \sim p(\cdot \mid \theta).$$

Here S_{obs} is the vector of summary statistics of the real record, $S(x)$ is the same function applied to a simulated one, and ε is the tolerance, or how far apart $d(\cdot, \cdot)$ is allowed to put them. Parameter values that reproduce the observation often have a high $A(\theta)$, and values that rarely do have a low one.

The second step reweights the prior by that acceptance probability:

$$p_\varepsilon(\theta \mid S_{\text{obs}}) \propto p(\theta) A(\theta).$$

A parameter value's weight in the approximate posterior is its prior weight multiplied by how often simulating at that value lands within ε of the observation. The subscript indicates that this is an approximation to the posterior, controlled by the tolerance ε. To keep things simple, in the code I never fix ε directly: I sort the simulations by distance and keep the closest 10%, which amounts to letting the data choose the ε that admits exactly 1,000 of them.

Three separate approximations are embedded in that construction, and keeping them distinct makes it much easier to diagnose a poor posterior. First, we compare $S(x)$ rather than x, which is exact only if S is sufficient for θ—in other words, only if the summary statistics capture the full impact of a given parameter value on the model's dynamics. Sufficiency essentially never holds in practice (Blum et al., 2013; Fearnhead & Prangle, 2012). Second, we accept a neighborhood rather than an exact match. As ε approaches zero we recover $p(\theta \mid S_{\text{obs}})$, but the acceptance rate approaches zero along with it. Third, the number of simulations is finite, so the posterior carries sampling error of its own.

The first matters most here, and is the one this tutorial is organized around: the other two shrink as simulations are added, whereas information discarded by the summary statistics cannot be recovered.

2.3 Why the number of summary statistics matters

The three methods are hurt to very different degrees by the first approximation—that the summary statistics may not be sufficient for the parameters. Rejection ABC reduces the comparison to a single Euclidean distance between summary statistics, and keeps a simulation when that distance is small. Being close on one statistic is not enough: a simulation has to be close on all of them at once, so matching on six is far more demanding than matching on one and the pool of qualifying simulations shrinks rapidly. Distances also concentrate: as dimension rises the nearest and furthest simulations become harder to tell apart, so the distance carries less information about which simulations actually resemble the observation. To keep enough simulations to build a posterior from, the tolerance has to be loosened, and loosened faster the more statistics there are. The 10% acceptance rate used here sounds like a tight match, but with six statistics it accepts a substantial share of the whole spread of simulated outcomes.

Rejection ABC should therefore give the widest intervals of the three methods, and Table 3 bears this out. An uninformative statistic still counts towards the match and forces the tolerance upward, so adding statistics can make inference worse rather than better. Anyone using rejection ABC must therefore choose a small number of them in advance, and the quality of the inference depends heavily on that choice.

ABC–RF removes most of that penalty. A random forest can decline to split on a statistic that carries no information, so an uninformative one costs little, and the method was developed partly to free researchers from reducing the dimensionality of their data in advance (Raynal et al., 2019). In practice this licenses summary vectors far larger than the number of parameters: an earlier study of frequency-dependent bias in music sampling used 178 summary statistics with `abcrf` to choose among three transmission models (Youngblood, 2019). However, the forest is still a regression on features the researcher has already computed: it can choose among the statistics it is given but cannot construct new ones, so the candidate set is still selected by hand.

NPE goes further. The network that compresses the data and the network that produces the posterior are trained together against the same objective, so the compression of the data into summary statistics is learned rather than assumed: the network is rewarded for retaining whatever predicts the parameters and discarding the rest. No distance is ever computed in the space of the data, so high-dimensional input carries no automatic penalty, and NPE can take in a full 37×6 time series directly where rejection ABC needs the data reduced to six numbers first.

3 THE MODEL AND DATA

3.1 The simulator

The simulator tracks a pool of active names over discrete years, and each year resamples the next cohort from the current one in the manner of a Wright–Fisher process (Wright, 1931). Innovation is conditioned on rather than modeled: new names enter at their observed counts, taken from an entrant schedule derived from the real record. With no innovation rate to estimate, the tutorial infers three parameters rather than four, a simplification I return to in Section 8. Each year proceeds with the following steps:

1. Names whose count has fallen to zero are removed from the pool.
2. Sampling weights are computed for the surviving names,

$$w_i \propto f_i^{\beta} \times \text{age}_i^{-\alpha} \times \exp(\eta_i), \quad \eta_i \sim \mathcal{N}(0, \gamma^2),$$

   where $f_i = \text{count}_i / \sum_j \text{count}_j$ is the current frequency of name i and age_i is its age in years since entry. A name's weight is the product of three terms: its current frequency, raised by the level of frequency bias (β); its age, raised by the level of age dependence ($-\alpha$); and a random multiplier $\exp(\eta_i)$, where the shock η_i is drawn fresh each year for each name. We write $\propto$ rather than an equals sign because only relative weights matter; they are normalized in the next step.
3. The births not claimed by entrants, $N_t - \sum_j \text{entrants}_j$, are allocated to the surviving names by multinomial sampling with probabilities $w_i / \sum_j w_j$, so that each remaining birth goes to a name drawn in proportion to its weight.
4. This year's entrants are appended to the pool at their observed counts with an age of 1, and the surviving names age by one year.
5. The six summary statistics are recorded.

Two details of this ordering are easy to get wrong. Entrants are appended after the multinomial draw, so a name in its first year cannot receive multinomial births and simply holds its observed entry count. Weights use each name's age before it is incremented.

The weight calculation, verbatim from `src/model.R`:

```
#compute sampling weights from frequency, age, and random
variation
freqs <- counts / sum(counts)
weights <- (freqs^beta) * (ages^(-alpha)) *
exp(rnorm(length(counts), 0, noise_sd))
probs <- weights / sum(weights)
```

The third argument of `rnorm` is a standard deviation, so γ is the standard deviation of the log-normal multiplier rather than its variance.

3.2 What each parameter does

β scales frequency dependence. At $\beta = 1$ a name's chance of being chosen is proportional to its current frequency: unbiased transmission, or the neutral model of cultural evolution (Bentley et al., 2004; Neiman, 1995). At $\beta > 1$ common names are favored beyond their frequency (i.e., conformity), which concentrates the distribution and tends to lower turnover. At $\beta < 1$ rare names are relatively favored (i.e., anticonformity), which flattens it. Because β is an exponent on frequency, the neutral point is 1 rather than 0, unlike models that parameterize bias additively—a common source of confusion when estimates are compared across studies.

α scales age dependence. The term is $\text{age}^{-\alpha}$, so positive values of α downweight old names and favor young ones (i.e., a novelty bias), while negative values favor established names. Setting $\alpha = 0$ removes the term.

γ scales random variation: as the standard deviation of a mean-zero normal in log space, it makes $\exp(\eta_i)$ log-normal with a median of 1. Small values perturb the weights slightly, while larger values may reorder them. Because the shock is redrawn independently each year for every name, γ mainly inflates the year-to-year variability of the summary statistics, and appears to have little effect on their long-run level; Section 6 argues that this distinction determines which methods can recover it. It is effectively a proxy for effective population size, since smaller populations produce exactly this kind of random year-to-year variation in frequencies. Setting $\gamma = 0$ also removes the only part of the model that lacks a tractable likelihood, since the weights are then fully determined by frequency and age, and the model could be fit by standard methods with no simulation at all. Some random variation is nonetheless needed for the other parameters to produce realistic cultural dynamics.

3.3 The six summary statistics

Each year the simulator records six numbers, comprising two diversity measures, two turnover rates and two measures of concentration.

Hill diversities (Chao et al., 2014; Hill, 1973) of order q are computed on the vector of name frequencies p_i:

$$D_q = \left(\sum_i p_i^q \right)^{1/(1-q)}, \quad D_1 = \exp\left(-\sum_i p_i \log p_i \right).$$

p_i is the share of the year's births going to name i, the f_i of the weight equation in Section 3.1, renamed here to match the diversity literature. The order q controls how much rare names count towards the answer, with higher values increasingly dominated by the commonest names, so D_q never rises as q rises and D_2 is always at most D_1. I use $q = 1$, the limiting case given by the second expression above and equal to the exponential of Shannon entropy, and $q = 2$, inverse Simpson. Both are measured in units of the effective number of equally common names, which makes them directly comparable and is the usual reason to prefer them to the raw indices (Roswell et al., 2021).

Turnover is calculated for the top k names, with $k = 10$ and $k = 100$, as the fraction of this year's top k that were not in last year's top k (Bentley et al., 2007; Evans & Giometto, 2011). Turnover is undefined in the first simulated year and is recorded as `NA`.

The two concentration measures are the proportion of births going to the single most popular name, and the proportion of births going to names used exactly once.

Each gives one number per year, and a method that cannot take a whole time series needs them collapsed. I collapse each in three ways: its median, its standard deviation and its lag-1 autocorrelation across years. The autocorrelation measures how strongly one year's value predicts the next, i.e., how persistent the trajectory is over time rather than how high it sits or how much it varies. Six statistics summarized three ways gives 18 features.

Including the standard deviation and the autocorrelation is a small departure from usual practice: applications of ABC in cultural evolution have often summarized a trajectory by its level alone (Youngblood,

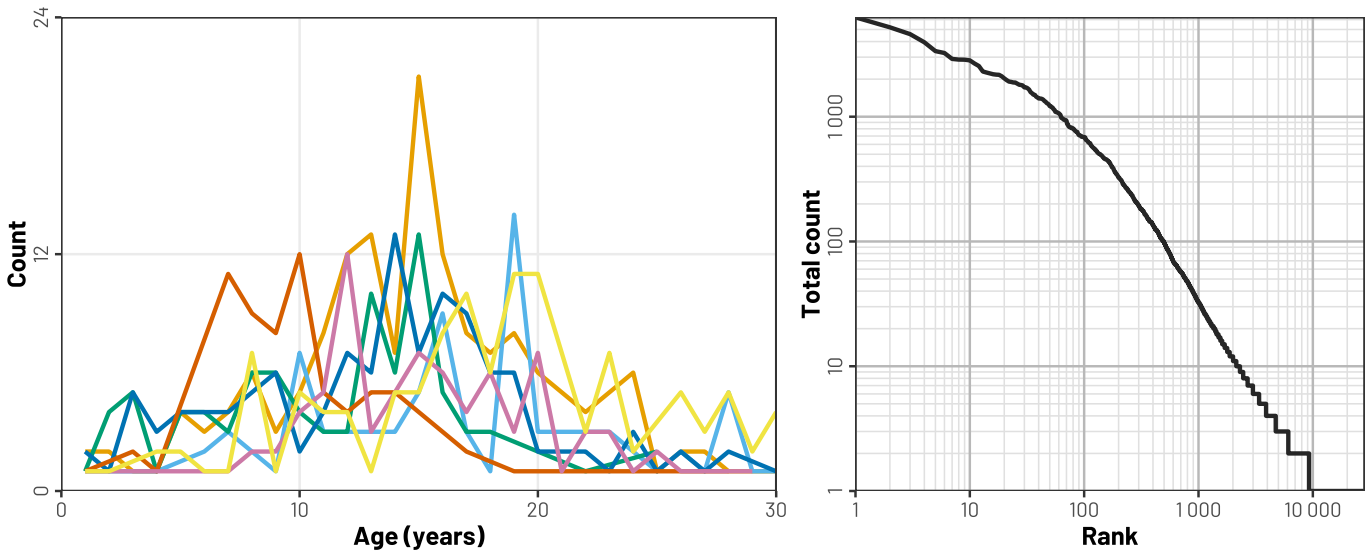


Figure 2: Left: example boom–bust name trajectories (count vs. name age). Right: log–log rank–frequency distribution of total name counts over the study period. Short-lived rises and falls motivate an age-dependent term in the model, and the heavy-tailed rank–frequency distribution is the aggregate pattern the simulator is asked to match.

2019). There are exceptions: (Crema et al., 2016) compare whole frequency trajectories rather than summaries of them, and (Kandler et al., 2017) argue that temporal change carries a stronger signature of the underlying process than any single snapshot. Section 6 gives my reason: a parameter that changes how a trajectory varies, rather than where it sits, leaves almost no trace in a median.

3.4 South Australian baby names

I use uncensored South Australian baby name counts (1944–2017), filtered to female names. Most published name data are censored, in that names falling below a threshold count are suppressed, and these are exactly the rare variants that carry much of the information about transmission (O'Dwyer & Kandler, 2017). I chose this register over the more commonly used national datasets because modeling that censoring correctly would add a layer between the data and the simulator that is irrelevant to the methods being compared. The simulation is initialized in 1980 from the observed name pool, with real name ages derived from the full record back to 1944, and run for the 37 subsequent years (1981–2017). Total births per year and the entrant schedule are taken from the observed data, so the simulator matches the record on cohort size and innovation by construction, and is asked only to reproduce the shape of the name distribution.

The observed record enters inference through the same six summary statistics. Computing observed and simulated statistics through different code paths is a common and easily missed source of bias, so both use `summary_stats()` in `src/model.R`.

Both the heavy-tailed structure and the short-lived rises and falls that motivate age-dependent weighting are visible in the record (Figure 2), although the simulator does not reproduce individual name paths.

4 THREE APPROACHES TO INVERTING THE SIMULATOR

All three methods share the same $M = 10^4$ prior simulations, drawn from

$$\beta \sim \mathcal{U}(0.5, 1.5), \quad \alpha \sim \mathcal{U}(-1, 1), \quad \gamma \sim \mathcal{U}(0, 0.5).$$

Here $\mathcal{U}$ denotes a uniform distribution, a random draw from anywhere between the two bounds. The ranges straddle the values that carry an interpretation, $\beta = 1$ for unbiased transmission and $\alpha = 0$ for no age effect, so that the prior does not commit in advance to a direction for either. The lower bound of 0 on γ is a constraint rather than a choice, since a standard deviation cannot be negative.

Simulating from the prior and overlaying the observed series is the cheapest check on whether these priors can produce data resembling the record, and it is always wise to run it before attempting inference (Figure 3). Section 7.1 reports it in full and takes up the turnover discrepancy it reveals.

The three methods form a ladder, each rung taking in more of the data than the one below (Table 2). Rejection ABC gets the medians alone because its acceptance region has to widen as statistics are added; ABC–RF can afford all 18 features because a forest can decline to split on a statistic that carries nothing; BayesFlow gets the full 37×6 series, from which any of those features could be reconstructed, and learns its own compression scheme. Summary statistics are z-scored to the training distribution (mean 0 and standard deviation 1 in the training set) before any distance is computed, so that statistics on different scales contribute comparably, and the observed statistics use the training mean and standard deviation rather than their own.

4.1 Rejection ABC

Rejection ABC is the oldest and most transparent of the three. Following the standard rejection algorithm (Beaumont, 2010), as previously applied to cultural data (Youngblood, 2019), inference was conducted with the following steps:

1. 10^4 iterations of the model were run to generate simulated summary statistics for parameter values drawn from the prior.
2. The Euclidean distance between the simulated and observed summary statistics was calculated for each iteration, after z-scoring both to the training distribution.

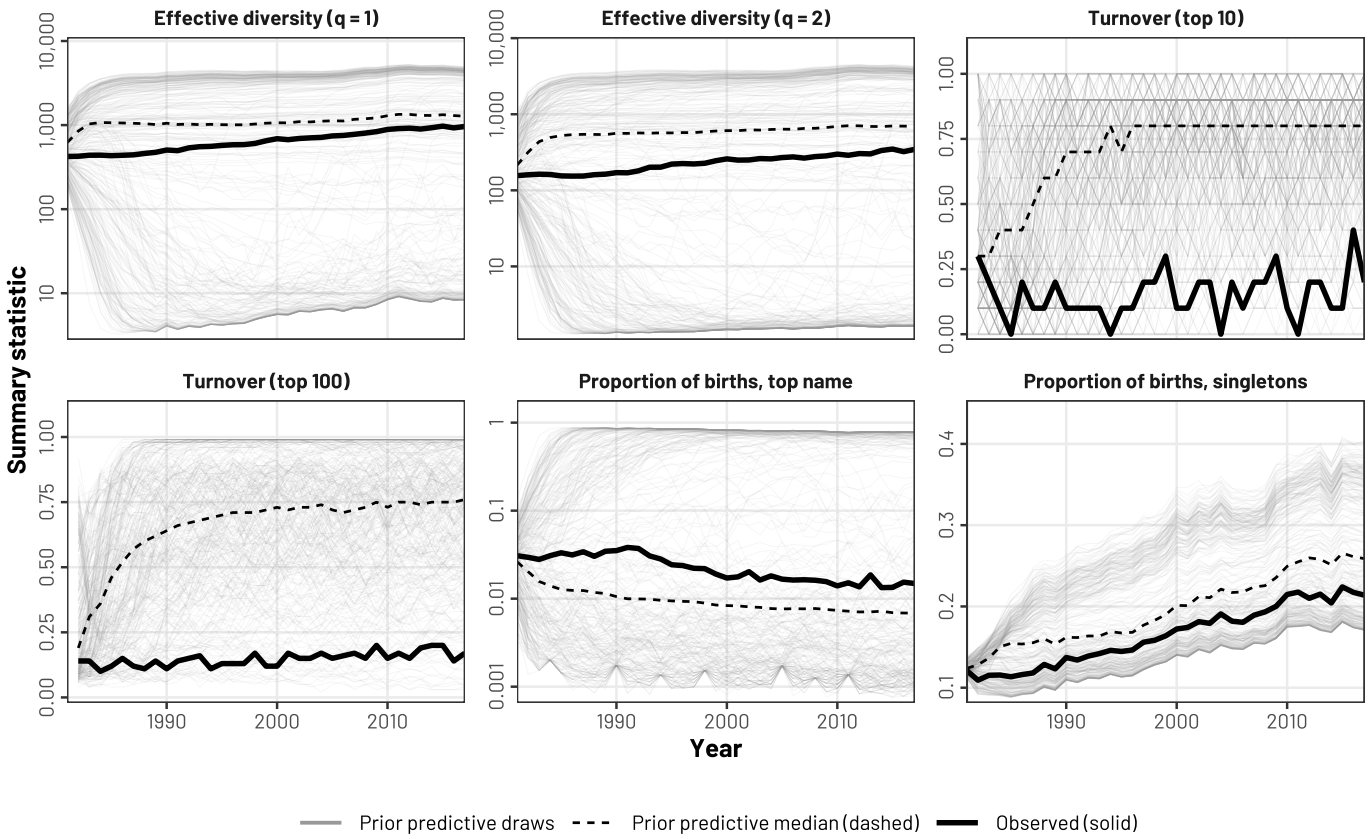


Figure 3: Prior predictive check. Each panel shows the yearly trajectories of one summary statistic across all 10^4 prior simulations, with the observed series overlaid in black. The diversity and concentration statistics sit comfortably inside the simulated envelope; both turnover statistics sit at its lower edge, indicating that the model produces far more year-to-year turnover than the record shows.

3. The 1,000 iterations with the smallest distances (i.e., a 10% acceptance rate) were used to construct the posterior distributions of β, α and γ.

In the code this amounts to three lines:

```
#rejection ABC on observed data: 1,000 closest of 10,000
(10% acceptance)
dist <- abc_distance(zscored$sim, zscored$obs)
best <- order(dist)[seq_len(n_accept)]
post_rejection <- prior[best, params]
```

The approach is simple to implement, has no tuning beyond the acceptance rate, and tends to fail visibly, since a posterior that resembles the prior is easy to spot. The costs are those described in Section 2.3, with one addition specific to the implementation above: every simulation outside the acceptance region is discarded, so 90% of the compute goes unused.

4.2 ABC with random forests

Rather than discarding simulations, ABC-RF uses all of them to learn a mapping from summary statistics to parameters. Random forest is a form of machine learning in which decision trees are trained on bootstrap samples of the data, with a random subset of predictors considered at each split, and averaged to predict an outcome (Raynal et al., 2019). Traditional ABC methods function best with fewer summary statistics (Blum et al., 2013), whereas random forests are more robust to their number and do not require potentially informative variables to be excluded (Raynal et al., 2019).

One regression forest of 500 trees was fitted per parameter using the `abcrf` package, with all 18 z-scored features as predictors:

```
#train one abc regression forest per parameter
stat_formula <- paste(STAT_NAMES, collapse = " + ")
train_data <- cbind(zscored$sim, prior[, params])
rf_models <- lapply(params, function(p){
  regAbcrf(
    as.formula(paste(p, "~", stat_formula)),
    data = train_data,
    ntree = 500, paral = TRUE, ncores = cores
  )
})
```

The forest does more than produce a point estimate. Each tree partitions the space of summary statistics, and two simulations falling in the same terminal leaf as the observation are, according to that tree, similar to it. Averaging leaf co-occurrence across trees assigns a weight to each training simulation, and resampling the training parameters by those weights yields posterior draws directly. I drew 5×10^4 such samples per parameter.

The approach has two limitations. Forests are fitted separately for each parameter, so joint structure in the posterior (e.g., a correlation between β and α) is not modeled. A forest can also only use information

present in the features it is given. The 18 features retain considerably more of each trajectory than the medians alone, but remain a compression chosen in advance, and anything they leave out—the shape of a year's frequency distribution beyond the six numbers taken from it, or temporal structure more elaborate than a lag-1 autocorrelation—is unavailable to the forest at any point.

4.3 Neural posterior estimation with BayesFlow

NPE takes a more direct route, and trains a neural network $q_\varphi(\theta \mid x)$ to output the posterior itself by minimising

$$\mathcal{L}(\varphi) = -\mathbb{E}_{\theta\sim p(\theta)}\mathbb{E}_{x\sim p(\cdot \mid \theta)}\left[\log q_\varphi(\theta \mid x)\right].$$

Here $q_\varphi(\theta \mid x)$ is the network's guess at the posterior and φ collects the network's weights, so training amounts to searching for the φ that minimises the loss $\mathcal{L}(\varphi)$. The two stacked expectations describe exactly the archive of simulations we already hold, and minimising the loss pushes the network's density onto the parameters that actually generated each simulation. Note that the letter q here denotes the network's density and is unrelated to the Hill order q of Section 3.3.

Because the expectations run over simulations we can generate at will, the loss is directly computable, and over a sufficiently flexible family of networks its minimiser is the posterior $p(\theta \mid x)$ itself. That is the theoretical reason to expect NPE to outperform methods that first compress x into hand-chosen summaries, though whether it does in practice likely depends on the architecture and the training budget. NPE also harnesses the capacity of modern deep learning, which makes it practical to fit models with more parameters and more complex structure than classical ABC can handle.

My implementation uses BayesFlow (Radev et al., 2022; 2023) and was conducted with the following steps:

1. The 10^4 simulations were split into a training set (90%) and a validation set (10%).
2. A summary network, implemented as a time-series transformer, was trained to compress each 37×6 trajectory into a learned 64-dimensional embedding.
3. An inference network was trained by flow matching (Lipman et al., 2023), a method that learns to turn draws from a simple base distribution into draws from the posterior, with the network conditioned on that embedding. Training ran for 500 epochs with a batch size of 128. BayesFlow builds a cosine learning-rate decay spanning the epoch budget, so the schedule is run to completion rather than stopped early.
4. The trained network was evaluated on the observed summary statistics to draw 1,000 posterior samples, and on the held-out test set described in Section 7.

Because the summary statistics are learned rather than chosen, the transformer determines what about the trajectory is worth retaining, reducing the influence of one of the researcher's most consequential modeling decisions. Inference is also amortized: training is slow and happens once, after which a posterior for a new dataset takes a single pass through the network. The obvious case for this is a workflow fitting the same model to many datasets, but a single dataset does not make the training a poor trade, because a single analysis is never a single model fit: the 200-simulation recovery test and 300-simulation calibration check reported in Section 7 both require inference on hundreds of simulated datasets, which are nearly free once the network exists.

BayesFlow is given the same six summary statistics described in Section 3.3, supplied as a time series rather than as medians, so the network learns how to compress a trajectory that I have already decided to compute; it never sees the raw output of the simulator, and the six statistics remain a hand-picked choice. I hold them fixed across all three methods so that the comparison isolates what each method does with the same information, but this understates what NPE can do. In a model of speed climbing, summary statistics were learned by a time-series transformer directly from the simulator's annual distribution of record times (Youngblood & Passmore, 2026); in a model of Mourning Warbler song learning, sixteen were learned from raw model output by a hierarchical deep-set and time-series transformer (Pitocchelli et al., 2025). Applied that way, the researcher decides only what the model should output, and the network decides which features to measure.

This comes at a cost. Training requires a Python toolchain, and a GPU or Apple Silicon device shortens it but is not a requirement: the network here is small and the training set modest, so the same schedule runs on an ordinary CPU several times more slowly—an overnight job rather than an impractical one.

The more serious cost is that the failure modes are harder to see. Rejection ABC reports on itself: there is a tolerance and an acceptance rate to look at, the accepted simulations can be inspected individually, and a run that has learned nothing produces a prior-like posterior a reader is likely to notice. A neural network offers no equivalent readout. It returns a posterior every time, and a network that trained badly (too few epochs, a summary embedding too small to hold the relevant signal, a learning rate that stalled) returns a narrow posterior centered on the wrong value, which on the page is indistinguishable from one centered on the right value. The loss curve will not separate them either, because it has no absolute scale: a curve that has flattened out says only that the network has stopped improving, not that it stopped in the right place. The check therefore has to come from outside the fit, which is why I treat the diagnostics in Section 7 as required rather than optional. The network is also only as good as the prior it was trained on, and may extrapolate without warning when evaluated far outside that region.

5 RESULTS

On the observed South Australian data, all three methods place β below 1, at 0.966, 0.957 and 0.953 (Figure 4, Table 3). The BayesFlow interval excludes $\beta = 1$ by a clear margin, the ABC–RF interval reaches just past it, and the much wider rejection ABC interval includes it comfortably. The record therefore leans anticonformist, with rare names adopted somewhat more often than their current frequency alone would predict. That conclusion rests on the sharpest of the three posteriors, though the calibration check in Section 7.3 finds no evidence that it is overconfident. A β below 1 is the same direction of effect as the negative frequency dependence (Newberry & Plotkin, 2022) report for naming records in four other countries, so this analysis is in effect a small-scale replication of that finding on an uncensored single-register dataset, using a simpler model that estimates one exponent rather than a nonparametric selection curve across frequency bins.

The methods agree less closely on α and γ. BayesFlow yields sharp posteriors, whereas rejection ABC remains comparatively diffuse, and for γ is nearly as wide as the prior, which indicates that it has extracted

Table 2: The three methods at a glance.

	Rejection ABC	ABC–RF	BayesFlow
Input	6 medians	18 features	37 × 6 series
Summary statistics	chosen	chosen	learned
Tuning	acceptance rate	`ntree`	architecture, epochs
Uses all simulations	no (10%)	yes	yes
Joint posterior	yes	no	yes
Amortized	no	no	yes
Toolchain	base R	R (`abcrf`)	Python (`bayesflow`)

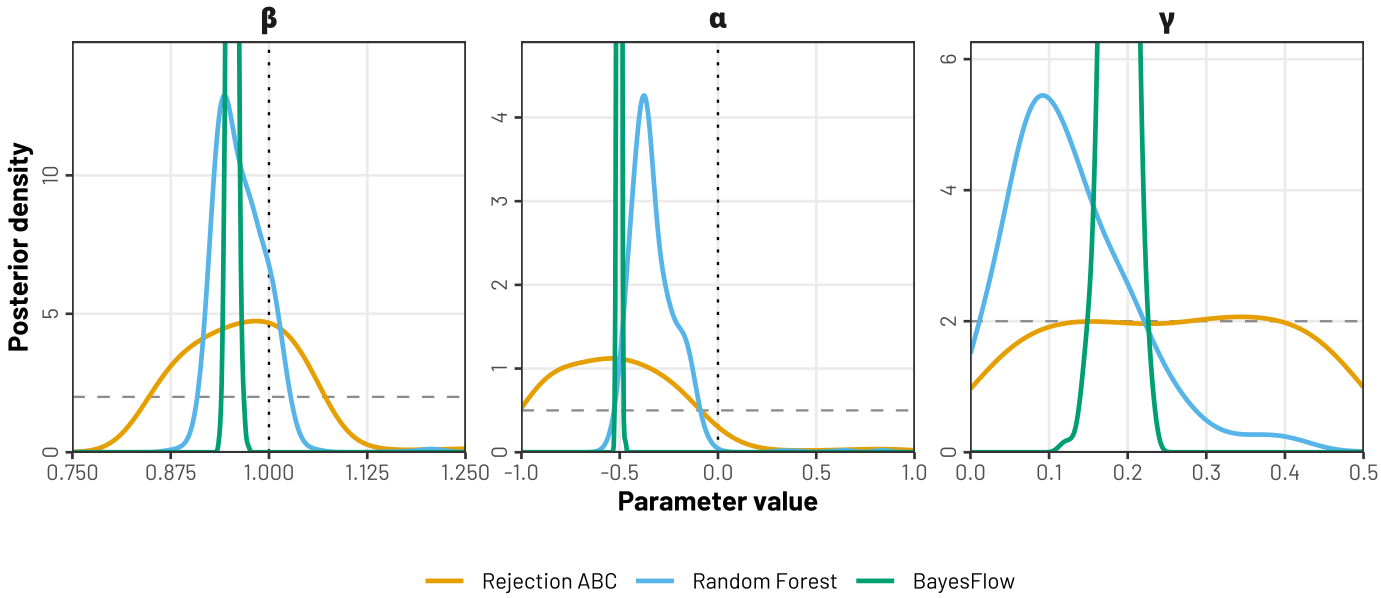


Figure 4: Posterior distributions for β, α and γ under rejection ABC (orange), ABC–RF (blue) and BayesFlow (green). Dotted vertical lines mark neutral frequency dependence ($\beta = 1$) and no age-dependent bias ($\alpha = 0$); the dashed horizontal line is the uniform prior. The β panel is clipped to (0.75, 1.25) rather than the full prior range of (0.5, 1.5), so that the three posteriors are distinguishable; the rejection ABC density continues beyond both edges. Each panel's vertical axis is also clipped just above the tallest non-neural density, so the much narrower BayesFlow peaks run off the top rather than flattening the other two.

Table 3: Observed-data posterior summaries, recovery correlations (Pearson r between true values and posterior means), and root-mean-square error as a multiple of the standard deviation of the true values, on a held-out test set of 200 simulations drawn from the full prior of Table 1, independently of the training simulations. RMSE/SD above 1 indicates an estimator that would do better returning the mean of the true values in the test set than using the data.

	Rejection ABC	ABC–RF	BayesFlow
β median (95% CI)	0.966 (0.846, 1.128)	0.957 (0.918, 1.021)	0.953 (0.944, 0.963)
α median (95% CI)	−0.531 (−0.977, 0.188)	−0.363 (−0.495, −0.074)	−0.502 (−0.515, −0.489)
γ median (95% CI)	0.255 (0.010, 0.488)	0.115 (0.010, 0.350)	0.189 (0.152, 0.219)
recovery r (β, α, γ)	0.96, 0.86, 0.39	1.00, 0.99, 0.94	1.00, 1.00, 0.97
RMSE / SD (β, α, γ)	0.28, 0.51, 0.95	0.09, 0.15, 0.35	0.05, 0.07, 0.24

very little information about that parameter. The ordering of interval widths is the same for every parameter—rejection ABC widest, then ABC-RF, then BayesFlow—and the geometry of the acceptance region predicts it, independent of the data.

ABC-RF and BayesFlow both place α firmly negative, at around −0.5, which means that weight increases with name age. This is a bias towards established rather than novel names, and is the opposite of the novelty bias the parameter was introduced to capture. The rejection ABC interval for α includes zero. These negative estimates may partly absorb misspecification rather than measure an age bias. New names are injected at their observed counts and cannot gain weight through α, so the age term is the only route by which the model can match the observed persistence of established names. The prior predictive check in Section 7.1 shows that persistence is exactly where the model struggles.

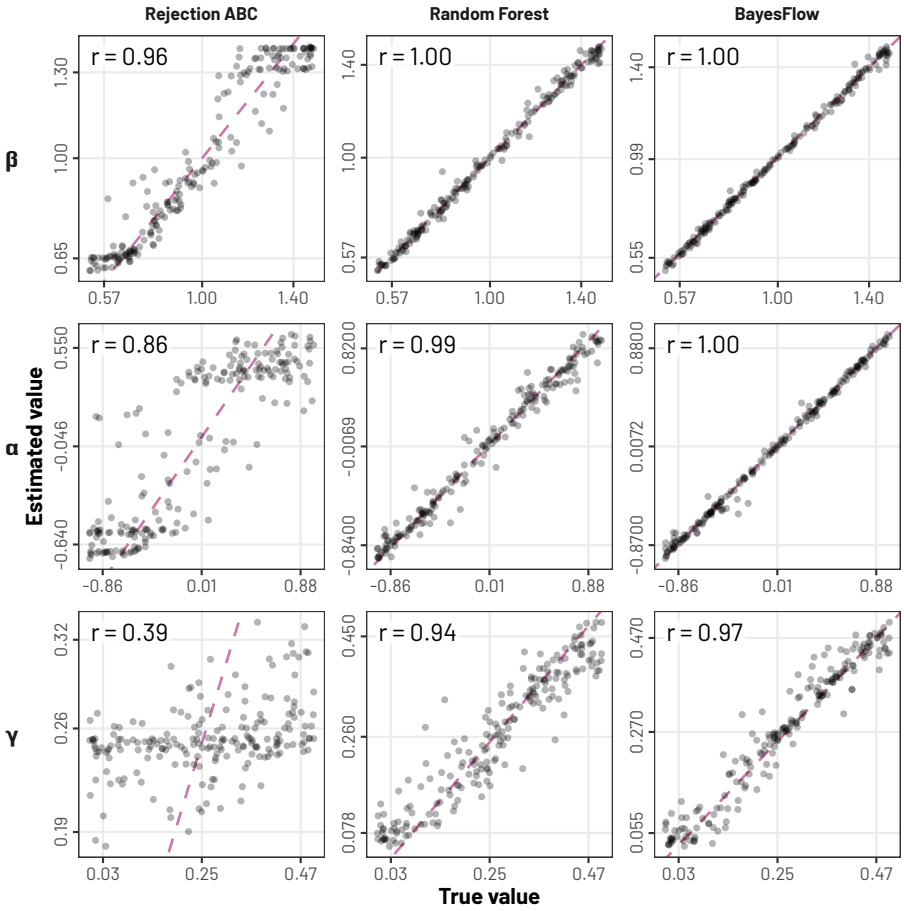


Figure 5: Parameter recovery on a held-out test set of 200 simulations drawn from the full prior, independently of the training simulations. Each panel compares true values to posterior means for one parameter and method, with axes scaled to each panel's own points, and the dashed line is the 1:1 identity. Annotated values are Pearson correlations.

Recovery on held-out parameters improves monotonically up the ladder, for every parameter (Figure 5). Both β and α are identified by the six medians alone, and rejection ABC reaches $r = 0.96$ and 0.86. It recovers γ poorly ($r = 0.39$), with an error nearly as large as the spread of the true values (RMSE/SD $= 0.95$), so on that parameter it barely improves on ignoring the data and returning the mean. ABC-RF, given the same six statistics plus their standard deviations and autocorrelations, recovers all three well ($r = 1.00$, 0.99 and 0.94), and BayesFlow, given the whole series, is better again ($r = 1.00$, 1.00 and 0.97).

BayesFlow has the greatest statistical power of the three, holding its advantage where the hand-built statistics serve worst, reaching $r = 0.97$ for γ with the lowest error of any method. The gap between rejection ABC and ABC-RF is nonetheless far larger than the gap between ABC-RF and BayesFlow, and twelve hand-computed features close most of the distance to a network trained on 222 numbers. A reader able to train an amortized network should expect the most power from it, though if the summary statistics are discarding information, fixing that is the more productive place to start.

6 WHY THE METHODS DIFFER

Table 3 could be read as evidence that the neural method is simply better. The gradient across parameters instead appears to track how much information about each parameter reaches the estimator.

Start with the median, which is the whole of what rejection ABC sees and the block that ABC-RF shares with it. The median of a sequence does not depend on the order of its elements, so any feature of a trajectory that depends on ordering (e.g., trend, autocorrelation, or the timing of a shock) is removed by taking the median, and no downstream estimator can put it back. The median is also a measure of location, so it discards most information about scale. The sequences (2.0, 2.1, 2.0, 1.9, 2.0) and (0.5, 3.6, 2.0, 0.9, 3.2) share a median of 2.0 but differ in standard deviation by a factor of roughly twenty.

The three methods are therefore not handed the same input (Section 4). Each rung of the ladder restores something the median removed, and BayesFlow is the only method not limited to features chosen in advance by the researcher.

The consequences of this depend on what each parameter does to a trajectory. β and α change the level of diversity, turnover and concentration, and raising β concentrates the distribution so that D_1 (diversity) falls and stays low. That is a shift in location, which the median tracks well. γ is a mean-zero shock redrawn every year, and much of what it does is change how a trajectory varies from year to year, which is the axis the median collapses.

Temporal summaries of this kind are often overlooked in cultural evolution, so I computed them here and then, as a teaching exercise,

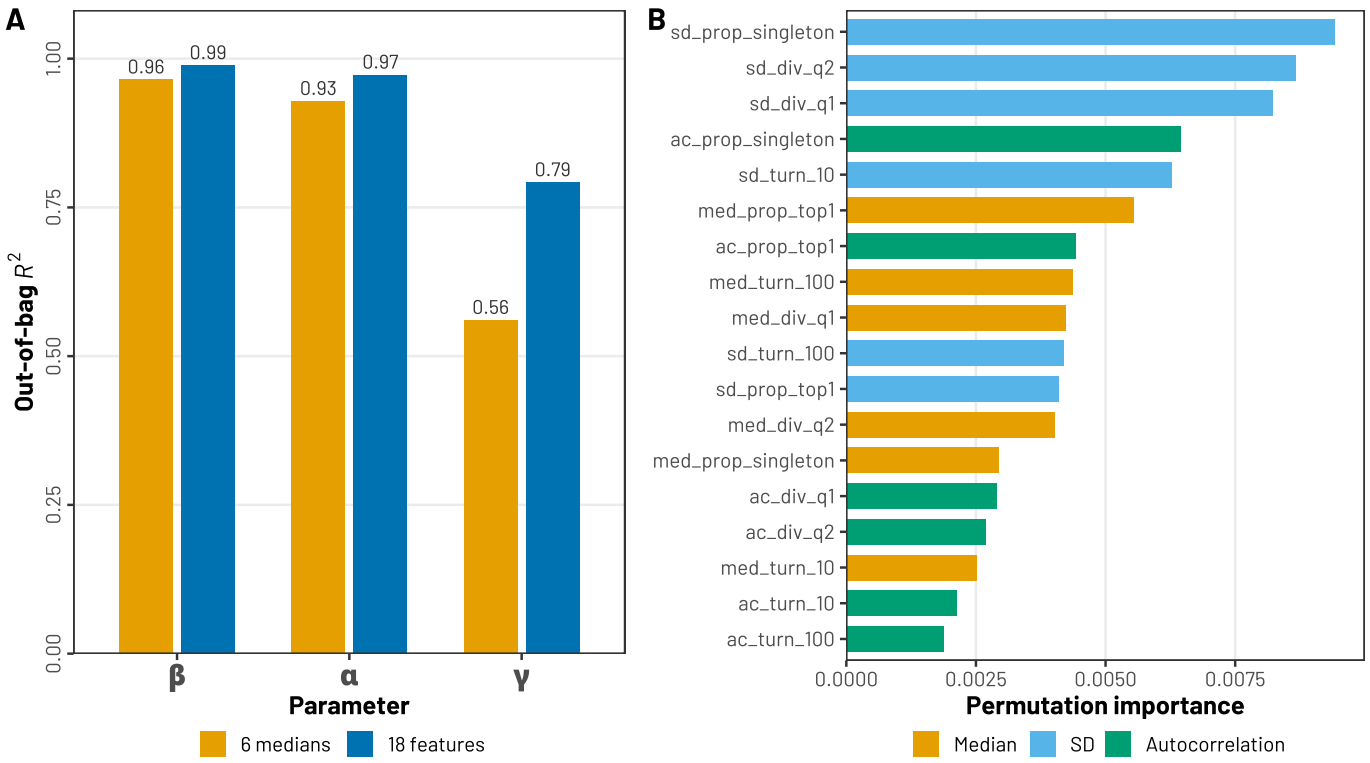


Figure 6: Where each parameter's signal lives. (**A**) Out-of-bag R^2 from random forests predicting each parameter from the six median statistics alone, and from all 18 features (median, standard deviation across years, and lag-1 autocorrelation of each statistic). Adding the temporal features is worth little for β and α and a great deal for γ. (**B**) Permutation importance in the 18-feature γ forest, colored by block. Standard deviations dominate, though no individual feature correlates with γ at $|r| > 0.12$.

compared a median-only fit against one that uses them (Figure 6). Fitting random forests to the same 10^4 simulations, the out-of-bag R^2 (the share of variance a forest explains on the simulations each tree was not trained on) rises only from 0.965 to 0.989 for β and from 0.929 to 0.973 for α when the twelve temporal features are added. For γ it rises from 0.561 to 0.792, and the standard-deviation block alone carries 48% of the permutation importance in that forest (how much accuracy drops when one feature is shuffled), with the autocorrelation block a further 24% against 28% for the medians.

None of this signal is visible marginally. Every one of the 18 features correlates with γ at $|r| < 0.12$, including the standard deviations the forest relies on most, whereas the strongest marginal correlations reach 0.83 for β and 0.71 for α. A researcher screening candidate statistics by their correlation with the parameter would have discarded every feature that works, because the signal for γ is conditional and interaction-driven, available to a learner that can condition on the other statistics and invisible to any pairwise screen.

The same contrast appears in a longstanding disagreement in this literature. Early work found the frequency distribution of popular names to be broadly consistent with drift (Bentley et al., 2004; Hahn & Bentley, 2003), using summaries of the standing distribution rather than its dynamics. Fitting a dynamical model to the annual time series instead, (Newberry & Plotkin, 2022) rejected neutrality decisively, and noted that the stationary distribution of name frequencies had not been able to. Both analyses used comparable records, and appear to differ mainly in which features of those records the inference procedure was given, which is the contrast this tutorial reproduces on a smaller scale.

More generally, the advantage of a method grows as a parameter's signal moves out of the level of the summary statistics and into their structure. Before choosing a method, then, ask what each parameter does to a simulated trajectory. If it shifts the trajectory up or down, median statistics will probably work, and if it changes how the trajectory varies, they probably will not. The same question indicates where effort is best spent, since for many problems the main constraint is the summary statistics rather than the inference framework.

7 DIAGNOSTICS

Posterior plots on real data cannot on their own indicate whether a method has worked, since a method that has learned nothing will still return a posterior. The following three checks help to separate the two cases.

7.1 Prior predictive checks

Before fitting anything, simulate from the prior and plot the resulting summary statistics against the observed ones. If the observed data fall outside the cloud of simulations, then no inference procedure can help, because either the model cannot produce data like the real data or the prior excludes the region in which it could, and both are modeling problems rather than inference problems.

It costs nothing, since the simulations already exist, and is probably the most informative of the three relative to the effort involved. I report mine here (Figure 3; numbers from `scripts/05_diagnostics.R`).

The observed medians of both Hill diversities, the top-name share and the singleton share all fall near the middle of the prior predictive distri-

bution, so the model handles the shape of the frequency distribution well. The two turnover statistics sit at the low edge of it, since the model produces more turnover than the record shows. All six observed values nonetheless lie inside their marginal 95% prior predictive intervals, so the posteriors are interpolation rather than extrapolation, and the difficulty in matching persistence bears on the interpretation of α below.

The check also shows how well the prior spends the simulation budget, and roughly two-fifths of it goes to degenerate runs the data exclude on sight, in which one name takes more than a tenth of all births or effective diversity falls below 50 names, almost all of them at $\beta > 1$.

7.2 Parameter recovery

Recovery involves simulating datasets from known parameters, running the full inference procedure, and comparing the estimates to the truth (Figure 5). It produced the bottom two rows of Table 3, and revealed that rejection ABC, given the medians alone, recovers γ poorly. Without it, the γ posteriors in Figure 4 look unremarkable, since rejection ABC returns a distribution over a plausible range and nothing about it indicates how little the data have moved it.

Recovery is best run over the full prior, and ideally before fitting to real data, since it establishes whether the summary statistics and the method can identify the parameters at all across the range you are prepared to entertain. Restricting the range of true values lowers correlations mechanically, which is a further reason to test the whole prior rather than a narrow window around the estimates you expect. Test parameters must be drawn independently of the training parameters, which is easy to get wrong when both come from the same seeded stream.

A correlation is in any case only interpretable relative to the range it was computed over, so I report root-mean-square error alongside r in Table 3. RMSE as a multiple of the standard deviation of the truth is the more decision-relevant number, since a value above 1 means the estimator would have done better ignoring the data. Rejection ABC approaches 1 for γ, at 0.95, so its γ estimates are barely more informative than the mean of the true values—a weakness the correlation alone understates.

7.3 Simulation-based calibration

Recovery checks the point estimate, whereas calibration checks the posterior as a whole. Simulate a dataset from a parameter value you already know, run the inference on it, and ask where that known value sits among the posterior draws; its position in that ordering is its rank. Repeat with a fresh known value each time. If the posteriors are right, the true value is an ordinary member of the crowd of draws, so its rank should be equally likely to land anywhere, and across many repetitions the ranks should come out uniform (Modrák et al., 2025; Talts et al., 2018).

Departures from uniformity are readable: ranks piling up at both ends indicate posteriors that are too narrow, ranks piling up in the centre indicate posteriors that are too wide, and ranks drifting towards one end indicate a posterior sitting systematically to one side of the truth. This matters most for the neural method, for the reason given in Section 4.3.

I ran the check on 300 held-out validation simulations, drawing 500 posterior samples from each (Figure 7). The rank histograms are consistent with uniformity for all three parameters, with chi-square p-values of 0.94 for β, 0.67 for α and 0.86 for γ on 19 degrees of freedom, and mean rank fractions near the expected 0.5. One of the 60 histogram bins, in the α panel, falls just below the pointwise band, which is fewer excursions than chance alone would produce. The rank ECDFs stay inside the simultaneous 95% band at every point, and Kolmogorov–Smirnov tests give $p = 0.83$, 0.34 and 0.31.

Figure 7 shows the same 300 ranks as binned histograms and as the departure of the rank ECDF from uniformity. A dip in the middle of a histogram with towers at either edge would signal intervals that are too narrow. Here the bars keep to the band, and in all three ECDF panels the line wanders and comes back without leaving the envelope.

The sharp BayesFlow intervals in Table 3 are therefore calibrated rather than overconfident, and the comparison of interval widths in Section 5 depends on that finding. The check does not prove the posteriors are correct, and with 300 replications it has limited power against small distortions, but overconfidence is the specific failure it is best at catching and the one that narrow intervals invite. Of the two views, the ECDF with a simultaneous band is the safer check, and the histogram the more familiar one.

8 COMMON FAILURE MODES

Most SBI analyses that go wrong do so in one of a handful of ways, and several of these are present, or nearly present, in this tutorial.

Summary statistics may not identify the parameter, which is what the γ result illustrates, and which is diagnosed by recovery rather than by posterior plots. Additionally, misspecification may be absorbed into a parameter. When a model cannot reproduce a feature of the data through the mechanism the modeler intended, the fitting procedure will often reproduce it through some other parameter, which then carries an estimate that is both precise and misleading (Frazier et al., 2020). The negative estimate of α reported above is a candidate, for the reason given in Section 5. Posterior predictive checks, in which data are simulated from the posterior and compared to the real data on statistics that were not used in fitting, are the usual way of detecting this.

Equifinality is harder to rule out, since different parameter combinations, and different models entirely, can produce very similar summary statistics (Barrett, 2019; Kandler et al., 2017). A related trap is that many inference procedures assume the population is at equilibrium, an assumption that is rarely warranted for cultural data and that can lead to incorrect conclusions when it fails (Crema et al., 2016; Kandler & Shennan, 2013). A tight posterior is evidence that the statistics identify the parameter within the assumed model, and says little about whether that model is correct. Priors that do not cover the data are a related concern, since if the observed statistics fall outside the simulated cloud, rejection ABC will still return its closest 10% and a neural network will still return a posterior. Both are extrapolations, and the prior predictive check described in Section 7.1 will usually reveal them.

9 DISCUSSION

Based on the results of all three methods, the female South Australian naming record between 1981 and 2017 leans anticonformist. Every method places β below 1, and the sharpest of the three excludes neutrality by a clear margin. The effect is modest, and its agreement with the naming records of the United States, France, Norway and the Netherlands (Newberry & Plotkin, 2022) is some reassurance that it is

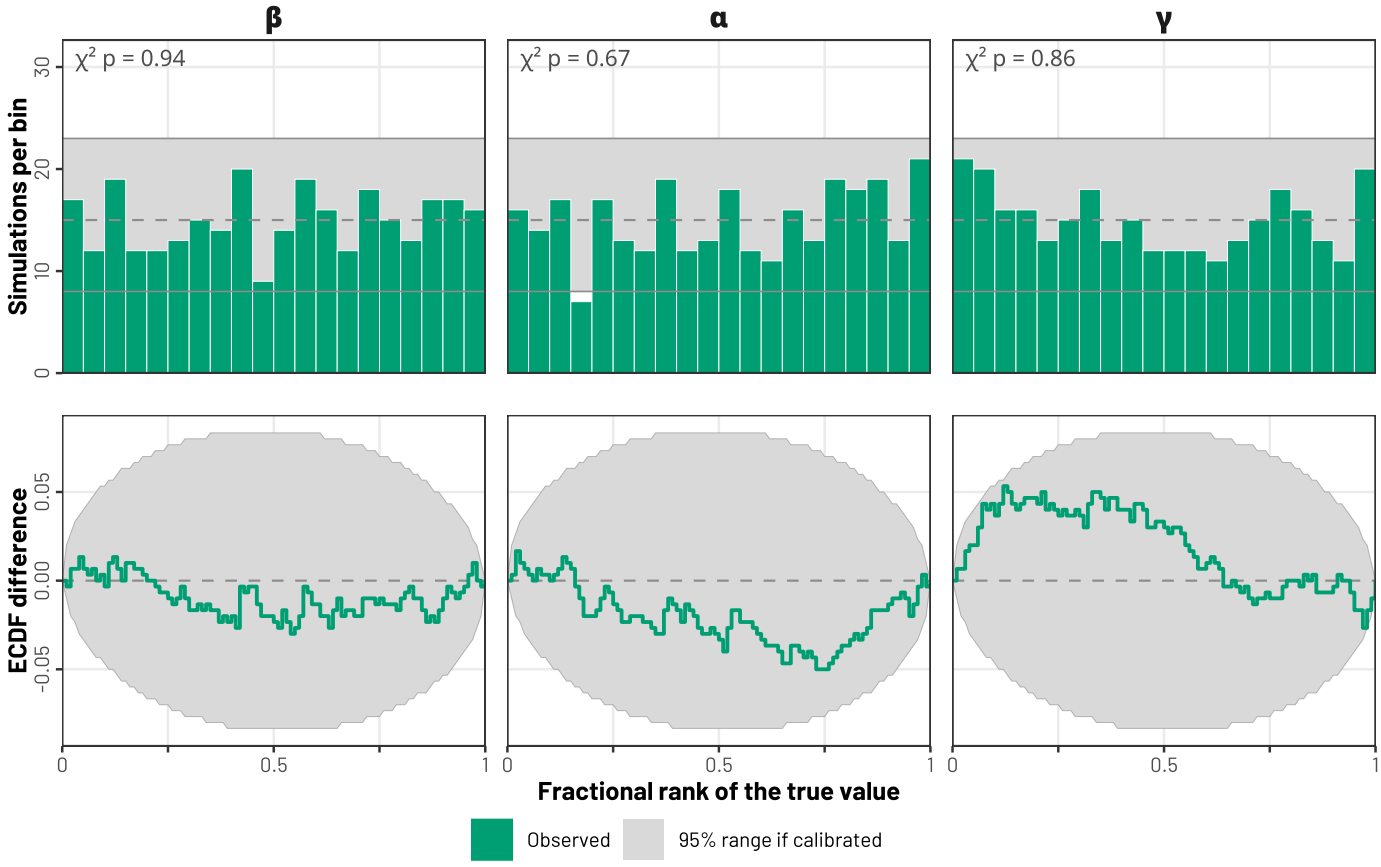


Figure 7: Simulation-based calibration of the BayesFlow posteriors, from 300 held-out simulations with 500 posterior draws each. Top: rank histograms of the true parameter value in 20 bins, with the dashed line marking the expected count of 15 and the grey band a pointwise 95% binomial interval. Bottom: the empirical CDF of the fractional ranks minus the uniform CDF, with a simultaneous 95% band. All three parameters stay inside the simultaneous band at every point, and 59 of the 60 histogram bins fall inside the pointwise band, consistent with calibration: a pointwise band of this kind is expected to be crossed by two or three bins in 60 through chance alone.

not an artefact of this register. I would still not lean hard on the magnitude, since β is correlated with α in the posterior and the age term may be absorbing persistence the model cannot otherwise produce.

This tutorial is a teaching tool rather than a definitive analysis of baby name dynamics. The model is deliberately close to a minimal cultural evolutionary account of naming, in which names are copied in proportion to how common they already are, with an age term and a random-variation term added. Models of roughly this form have a long history in the field as baselines against which richer accounts are compared (Bentley et al., 2004; Hahn & Bentley, 2003; Neiman, 1995), and their value lies in being simple enough that the consequences of each assumption can be traced, which is also what makes the model useful for teaching. Students can run the full pipeline, see where the three methods agree, and see why neural amortization helps once the summary statistics are high-dimensional time series rather than a handful of medians.

Several limitations follow from the design. Innovation is conditioned on rather than inferred, so the tutorial cannot speak to how new names arise, and this is also the most likely source of the misspecification discussed above. Median statistics discard temporal information by construction, which accounts for the γ result under rejection ABC. The simulator reproduces aggregate patterns but not individual name paths. Results are also for female names in a single uncensored register, and the parameter estimates should not be read as general facts about naming.

Three practical points may generalize beyond this example. First, where a study has the simulations to train on and the question turns on a parameter that is hard to summarize, setting up an amortized network repays the effort: neural posterior estimation had the greatest statistical power of the three here, its advantage was largest for the parameter the hand-built statistics served worst, and simulation-based calibration found no evidence that its narrow intervals are overconfident. Second, ABC–RF occupies a useful middle ground, fast to train in R, yielding genuine posterior weights, tolerating uninformative summary statistics and coming close to BayesFlow here once the temporal features were included, although it cannot recover what the summaries have already discarded. Third, deciding what each parameter does to a trajectory before settling on an inference method is a useful habit in its own right, and doing so here would have anticipated most of the results reported.

A final note concerns a tool many readers will now be using in some form. Large language models are helpful for the inference half of an SBI workflow. The ABC, forest and BayesFlow code in this repository is largely scaffolding around well-documented and reasonably stable

Table 4: The pipeline, step by step. Steps 1 and 2 take a few minutes each on $N - 1$ cores, the random forests dominating step 2; steps 4 and 5, seconds. Step 3 takes roughly 100 minutes on an Apple GPU for the full 500-epoch schedule. `QUICK_RUN = True` gives a 10-epoch pass fast enough for a classroom session, though it reproduces only the ordering of the results; `RESTORE_CHECKPOINT = True` instead restores the trained network shipped in `output/bayesflow_checkpoints/`, skipping training.

Step and script	Writes to `output/`	Produces
1. Simulate `scripts/01_simulate.R`	`prepared_data.rds` `sbi_results.rds`	The shared archive of 10^4 simulations used by all three methods

2. Approximate Bayesian computation
`scripts/02_abc.R`
	`abc_posteriors.rds`
	`recovery_testset.rds`
	`recovery_estimates.rds`
	`abc_inference_results.txt`
		Rejection ABC and ABC–RF columns of Table 3; the shared 200-simulation recovery test set

3. Neural posterior estimation
`scripts/03_bayesflow.ipynb`
	`bayesflow_posterior.csv`
	`recovery_bayesflow.csv`
	`bayesflow_sbc_ranks.csv`
		BayesFlow column of Table 3; diagnostics in Sections 7.2 and 7.3, from the notebook's recovery and calibration cells

4. Figures `scripts/04_figures.R`	`figures/` `inference_results.txt`	Figure 2, Figure 4, Figure 5, and the summary behind Table 3
5. Diagnostics `scripts/05_diagnostics.R`	`diagnostics.txt` `figures/`	Figure 3 and the numbers in Section 7.1, Figure 6 and the forest comparison in Section 6, Figure 7 from step 3' s ranks

interfaces, and mistakes in it tend to announce themselves as errors or as diagnostics that fail. The simulation half is another matter. In my experience these tools still struggle to build simulation models that are properly grounded in the existing literature. They readily produce a simulator that runs and looks reasonable while quietly substituting a plausible mechanism for the one a particular body of work specifies, or adopting a parameterization that does not match the papers being built on, which makes the estimates incomparable to published ones. The diagnostics offer little protection, since recovery and calibration ask whether inference works for the model as written rather than whether that model is the one intended, and a substituted mechanism that still reproduces the observed summary statistics will pass the prior predictive check as well. If you are using an AI-assisted coding workflow, I would recommend only using these tools for the inference code, and writing the simulator yourself (or at minimum read it line by line against the sources it claims to implement).

10 REPLICATING THE ANALYSIS

Everything reported here can be reproduced from a clean copy of the repository (URL to be added). There is no single command that runs everything; the steps below are meant to be run one at a time, since each writes intermediate files worth a look. All generated files land in `output/`, and all paths below are relative to the project root.

Before starting, install the R dependencies (`here`, `hillR`, `abcrf`, `ranger`, `ggplot2`, `ggh4x`, `cowplot`, `showtext`, `svglite`, `scales`) and the Python dependencies listed in `requirements.txt`, which require BayesFlow 2.x. The most useful file to read first is `src/model.R`, which holds the simulator and the six summary statistics with no configuration machinery in between, and whose `STAT_NAMES` fixes the order of the statistics used everywhere downstream.

Step 1 loads `data/southaustralia_babynames_19442017.csv`, filters to female names, builds the 1980 initial name pool with real name ages taken from the full record back to 1944, and derives the per-year entrant schedule. It then draws 10,000 parameter sets from the priors in Section 4 and runs the simulator once for each. Steps 2 and 3 fit the three methods to it; steps 4 and 5 produce the figures and diagnostics above.

11 ACKNOWLEDGEMENTS

I would like to thank Anne Kandler, Enrico Crema, Adam Powell, Simon Carrignon, Robert Lachlan, and Dominik Deffner for advancing the use of simulation-based inference in cultural evolution, and, in the case of some, inspiring me to explore these methods earlier in my career.

12 AUTHOR CONTRIBUTIONS

M.Y. conceived the study, wrote the code, conducted the analyses and wrote the manuscript.

13 FINANCIAL SUPPORT

This research received no financial support.

14 COMPETING INTERESTS

The author declares no competing interests.

15 DATA AVAILABILITY STATEMENT

The uncensored South Australian baby name records used in this study, together with all simulation output needed to reproduce the results, are available in the project repository (https://github.com/masonyoungblood/sbi_tutorial).

16 RESEARCH TRANSPARENCY AND REPRODUCIBILITY

All code for the simulator, the three inference methods, and every figure and diagnostic reported here is available in the project repository (https://github.com/masonyoungblood/sbi_tutorial). Section 10 describes how to rerun the full pipeline from a clean copy of the repository.

17 AI DECLARATION

M.Y. used AI tools in the process of editing this manuscript.

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